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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
FOURTH SEMESTER
Skill Enhancement Course (SEC) - MG4SECECT202, MG4SECECC200 -
SOLAR TECHNOLOGY AND APPLICATIONS (2024 ADMISSION ONWARDS)

SCHEME OF EVALUATION

Part A - MCQ

Answer all questions

Each question carries **1** mark

1. c). Ω
2. b). Early fault detection
3. a) Switch off and isolate the system
4. c) High
5. b) Excess heating
6. d) Converted into electrical energy
7. a) Maximum Power Point Tracking
8. b) Local employment generation
9. b) 3600 Kj
10. a) Overcharging of battery
11. a) Protect against electrical shock
12. c) Convert DC to AC
13. b) Specific yield
14. b) 0.5 A
15. d) Traction battery
16. c) Li-ion battery
17. c) Ampere-hour (Ah)
18. c) Excitons
19. c) Battery
20. a) Sunlight intensity
21. b) Lightweight and the ability to bend
22. d) Store excess energy
23. b) Faulty charge controller
24. d) Peak output power
25. c) Installation failure
26. c) DC motor
27. d) DC solar cable
28. b) Maximize energy generation

29. a) Electron–hole pairs
30. d) Non-mechanical device
31. d) Solar cell
32. d) AC output
33. b) Floating solar PV systems
34. a) Salt bridge
35. c) Both electrical and environmental data

Part B -Short Essay Questions

Answer any **3** questions Each question carries 5 marks

36. Thin-film solar photovoltaic (PV) -2

    Advantages: any three – 1.5

    Disadvantages: any three – 1.5

    Thin-film solar photovoltaic (PV) materials are semiconductor materials deposited in very thin layers onto substrates such as glass, metal, or plastic. Unlike conventional crystalline silicon solar cells, thin-film solar cells use much less semiconductor material, making them lightweight and flexible.

    Advantages: Lightweight, flexible, lower manufacturing cost, better performance in low light

    Disadvantages: Lower efficiency, larger installation area required, shorter lifespan, lower energy density

37. Any five safety precautions – 5 marks

    Use Personal Protective Equipment (PPE)

    Isolate electrical circuits

    Handle solar panels carefully

    Avoid working in adverse weather

    Ensure proper earthing (grounding)

    Use proper wiring practices

    Maintain safe working area

    Inspect roof strength

38. Brief explanation: 5 marks

    The inverter acts as the heart of a solar PV system, converting DC power into usable AC power, maximizing energy extraction through MPPT, ensuring safe operation, and

enabling efficient utilization of solar energy in homes, industries, and grid-connected applications.

39. Brief explanation: 5 marks

    Real-time monitoring enhances solar PV system performance by detecting faults early, improving efficiency, reducing maintenance costs, increasing energy generation, and enabling remote supervision. It plays a crucial role in ensuring the reliable and efficient operation of solar power systems.

40. Uses: 3 marks

    Advantages: 2 marks

    Tubular batteries are widely used in solar energy systems, inverters, UPS systems, and industrial backup applications because of their long lifespan, high cycle life, reliability, and excellent performance under deep discharge conditions. These features make them one of the most preferred batteries for energy storage applications.